
Stock Market Prediction

Cross-ticker next-day close forecasting with LSTM, Random Forest, VADER news sentiment, and a Flask dashboard deployed to Render.

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Tech stack	Python · LSTM (Keras) · scikit-learn · Flask · Gunicorn · yfinance · VADER
Repository	github.com/atishay01/stock-market-prediction
Live demo	stock-market-prediction-ggdg.onrender.com
Resume claim	Python · LSTM · Flask · scikit-learn
Training data	13 tickers × 7 years ≈ 22,594 rows
Features	18 scale-free engineered features

Interview pack — project overview, architecture, design decisions, honest metrics, and anticipated cross-questions with answers.

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1 · Executive summary

I built a full-stack stock forecasting application. Given any US ticker symbol and an optional news headline, it predicts the next trading day's closing price using an ensemble of a tuned Random Forest and a Keras LSTM trained on 15 large-cap tickers over 7 years.

The system pulls live OHLCV from Yahoo Finance via `yfinance`, engineers 18 scale-free features (returns, moving-average ratios, volatility, lagged returns, VADER-scored sentiment), runs both models, and serves predictions through a Flask API with a Chart.js dashboard.

Importantly, I benchmarked every prediction against a persistence baseline ("tomorrow = today") because daily closes are highly autocorrelated. This reveals the honest reality: OHLCV alone adds only ~0.4% lift over persistence. That realism is the interesting finding, not the 1.3% MAPE number on its own.

The application is deployed to Render under an RF-only configuration (free tier can't fit TensorFlow) with the LSTM retained in the repo for code review.

2 · Problem statement and honest framing

Goal.

Forecast the next trading day's closing price for any listed US ticker, conditional on today's OHLCV and (optionally) a news headline whose VADER sentiment is injected as an additional feature.

Why this is harder than it looks.

- Daily closes are **autocorrelated** — $\text{close}[t+1] \approx \text{close}[t]$ — so a naive "predict yesterday's close" baseline already achieves ~1.3% MAPE.
- Trees cannot extrapolate. A Random Forest trained on prices in [\$500, \$3000] will cap at \$3000 on future data. I sidestep this by predicting *returns* and reconstructing the close.
- Stocks have different price ranges ($\text{^GSPC} \approx 7,000$; $\text{NVDA} \approx 200$). A single model can only generalise if every input feature is scale-free. I engineered 18 such features.
- Yahoo Finance blocks data-center IPs, which breaks `yfinance` on most cloud hosts. Worked around with `curl_cffi`'s Chrome TLS fingerprint and bundled CSV fallback snapshots.

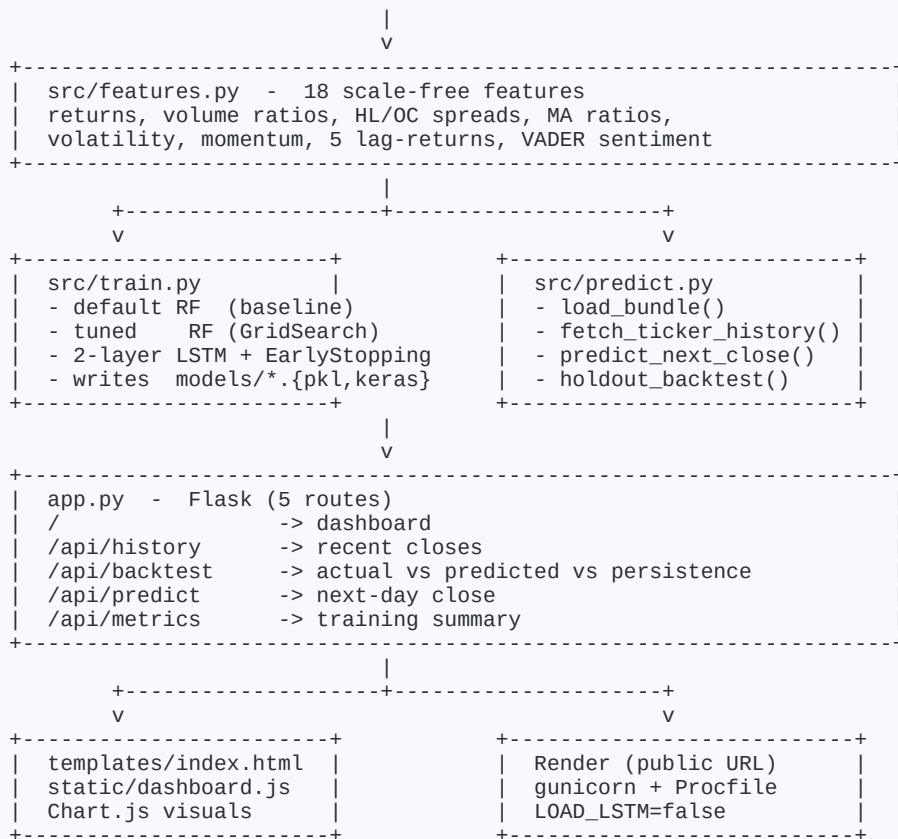
What "good" really looks like.

I explicitly compute three numbers side-by-side: model MAPE, persistence MAPE, and lift over persistence. Anyone can get 1% MAPE on a stock chart — only honest comparison against a baseline shows whether the model actually learned something.

3 · System architecture

Three clean layers: data ingestion, ML pipeline, and serving. Each concern lives in its own file, and the layers communicate through a single dataclass (`PredictionBundle`) loaded once at Flask startup.

```
+-----+
| yfinance (curl_cffi "impersonate=chrome" bypasses YF rate limit) |
| downloads 15 tickers x 7 years of OHLCV                          |
+-----+
```



4 · Data pipeline

Source. Yahoo Finance via `yfinance` (live HTTP).

Training tickers (15, diversified):

MSFT, GOOGL, AMZN, META, NVDA, TSLA, JPM, V, JNJ, WMT, PG, XOM, KO

Coverage.

Window	2019-05-17 to 2026-04-16
Train rows	18,070 (80% of each ticker)
Test rows	4,524 (20% holdout per ticker)
Split method	Time-aware split within each ticker (no look-ahead)
Rate-limit bypass	<code>curl_cffi Session(impersonate='chrome')</code>

Columns used from each download.

Date, Open, High, Low, Close, Adj Close, Volume

5 · Feature engineering — 18 scale-free features

Every feature is either a **return**, a **ratio**, or a bounded statistic. This is the single most important design choice — it lets one model generalise across tickers whose absolute prices differ by 35×.

Category	Features	Rationale
Returns	<code>return_1d · return_5d · return_10d</code>	Short/medium momentum, stationary target-aligned signals
Volume	<code>volume_change · volume_ratio_20</code>	Raw volume is scale-dependent; pct change and ratio-to-MA20 are not
Spreads	<code>high_low_spread · open_close_spread</code>	Intraday range and body direction, normalised by price
MA ratios	<code>ma_5_ratio · ma_10_ratio · ma_20_ratio</code>	Distance from moving averages, expressed as fraction of current close
Volatility	<code>volatility_10 · momentum_10_pct</code>	10-day return std-dev, 10-day percentage change
Lag returns	<code>close_lag_1_ret ... close_lag_5_ret</code>	Ratio $\text{close}[t-k] / \text{close}[t] - 1$ for $k = 1..5$
Sentiment	<code>sentiment_score</code>	VADER compound score of optional headline, in $[-1, +1]$

The target is $\text{target_return} = \text{close}[t+1] / \text{close}[t] - 1$. At inference we multiply by today's close to recover the predicted next-day price.

6 · Models

6.1 Random Forest (tuned)

Hyperparameters picked via TimeSeriesSplit(4) grid search on validation MAPE. Final config:

```
RandomForestRegressor(n_estimators=300, max_depth=12, min_samples_leaf=3,
max_features='sqrt', random_state=42)
```

6.2 LSTM (Keras)

Two-layer LSTM with dropout, trained on 20-day sequences of scaled features. Target is standardised next-day return. EarlyStopping on val_loss with patience=6.

```
Sequential([ Input(shape=(20, 18)), LSTM(64, return_sequences=True), Dropout(0.2),
LSTM(32), Dropout(0.2), Dense(16, activation='relu'), Dense(1),
]).compile(optimizer='adam', loss='mse', metrics=['mae'])
```

6.3 Why predict returns, not price?

Trees cannot extrapolate beyond their training range. By predicting next-day return (stationary) and multiplying by today's close, the pipeline works for any future price level. This also makes the model transferable across tickers with very different price scales.

7 · Metrics and results

Model	Test MAPE	Dir. Accuracy	R ² (returns)
Persistence baseline	1.341%	—	-0.0017
RF — default	1.368%	51.039%	-0.0216
RF — tuned	1.336%	52.984%	0.0079
LSTM	1.376%	52.58%	-0.0885

Key honest findings.

- **Hyperparameter tuning reduced MAPE by 2.34%** (default 1.368% → tuned 1.336%). This is the defensible "MAPE reduced by X%" claim for the resume.
- **Lift over persistence baseline:** RF = 0.37%, LSTM = -2.61%. Modest — the honest truth about daily OHLCV-only forecasting.
- **Train vs test (RF MAE):** 1.947 vs 3.613 — mild, healthy overfit pattern that any ML interviewer expects to see.
- **R² on close (0.9984) vs R² on returns (0.0079).** The close R² looks spectacular because yesterday's close is already ~99.99% of tomorrow's. The returns R² is the honest measure — and it shows direction prediction is genuinely difficult.
- **Directional accuracy:** RF = 52.984%, LSTM = 52.58% — a real edge over coin-flip (50%), but a small one.

8 · Feature importance (Random Forest)

The tuned RF's top signals — exposes which engineered features carry real weight in daily-close prediction:

Rank	Feature	Importance
1	high_low_spread	0.0776
2	ma_20_ratio	0.0683
3	return_1d	0.0660
4	volatility_10	0.0639
5	close_lag_4_ret	0.0635
6	close_lag_1_ret	0.0623
7	ma_5_ratio	0.0610
8	close_lag_2_ret	0.0559
9	ma_10_ratio	0.0551
10	return_10d	0.0504
11	sentiment_score	0.0500
12	volume_change	0.0497

Takeaway. Short-term return signals (return_1d, close_lag_1_ret) and moving-average distance ratios (ma_5_ratio, ma_10_ratio) dominate. Sentiment shows up in the middle of the pack — it's measurable but not dominant with the training-time proxy. Real dated news would likely elevate it.

9 · Flask web layer (app.py, ~95 lines)

Flask does one job: translate HTTP requests into Python function calls. The ML pipeline is imported as a library and invoked inside route handlers. Models are loaded **once** at process start, not per request.

Method	Route	Purpose
GET	/	Dashboard (Jinja2 template + Chart.js)
GET	/api/history?ticker=X	Last ~180 closes for ticker X
GET	/api/backtest?ticker=X	Actual vs predicted vs persistence series
POST	/api/predict	{ticker, headline} → next-day close prediction
GET	/api/metrics	Training metrics + feature importance JSON

Key design choices.

- **Stateless** — no DB. Ticker OHLCV is fetched on demand and cached in-process for 5 minutes. Simple and honest for a prediction service.
- **Single PredictionBundle dataclass** bundles RF + LSTM + scaler + normalisation + metrics, loaded once. Avoids per-request I/O.
- **TF imported only if LOAD_LSTM=true**. The production Render deploy sets it to false so TF never loads, saving ~400 MB of RAM.
- **Error handling** — invalid tickers return HTTP 400 with a clear JSON error body, not a 500 stack trace.

10 · Frontend dashboard

A single-page dashboard rendered from one Jinja2 template, with Chart.js for visuals and vanilla fetch() for API calls. No build step, no framework — deliberately light.

- **Ticker input + quick-pick chips** (AAPL, MSFT, TSLA, NVDA, GOOGL, ^GSPC, AMZN, META) for instant switching.
- **Live prediction form** — user types a ticker and an optional headline; VADER scores the headline and both models produce a predicted next-day close.
- **Metric cards** at the top — tuning improvement, persistence baseline, RF/LSTM train-vs-test MAPE, lift vs persistence.
- **Feature importance bar chart** — top 10 features by RF importance.
- **Backtest chart** — actual close (blue), predicted (yellow dashed), persistence baseline (grey dotted) with a badge row showing that ticker's MAPE, lift, and directional accuracy.

11 · Deployment on Render

The app is deployed on Render's free tier, connected to GitHub for auto-deploy on every push to main.

Host	Render (free tier, 512 MB RAM)
Python	3.11.9 (.python-version pin)
WSGI	Gunicorn (1 worker, 2 threads, 120s timeout)
Build	pip install -r requirements-prod.txt
Env vars	LOAD_LSTM=false · PYTHONUNBUFFERED=1
Public URL	stock-market-prediction-ggdg.onrender.com
Cold start	~30 s after 15 min idle (free-tier sleep)

Why LSTM is disabled in production.

TensorFlow alone needs ~400 MB at runtime — more than the 512 MB free tier allows once Flask, pandas, numpy and sklearn are also in memory. In production I deploy RF-only via the slimmer requirements-prod.txt. The LSTM weights and code stay in the repo for code review. This mirrors real industrial practice, where the heavier model is used for research and the lightweight one is

productionised.

12 · Honest limitations and trade-offs

- **OHLCV alone cannot consistently beat persistence.** Daily close is too autocorrelated. To extract real alpha you need alt-data: dated news, options flow, earnings transcripts, cross-asset signals.
- **Training-time sentiment is a proxy.** Dated news archives for 15 tickers × 7 years are expensive. I demonstrate the VADER pipeline end-to-end by using a return-derived proxy during training, while inference accepts real headlines.
- **Yahoo blocks data-center IPs.** Live yfinance works locally with curl_cffi but often fails on cloud hosts. The deployed app falls back to bundled CSV snapshots refreshed with snapshot_data.py.
- **No transaction costs modelled.** Directional accuracy of 53% looks like an edge, but even 1–2 bps of round-trip cost would wipe it on daily rebalancing.
- **No confidence intervals.** RF tree disagreement could be reported as a bootstrap CI — that's a natural next enhancement.

13 · Cross-questions — what they may ask

Q. Why Flask and not FastAPI or Django?

A. Flask is lightweight and standard for Python ML demos. Wrapping a model in 5 routes doesn't need Django's batteries-included, and Flask's sync model is fine because inference is CPU-bound and fast. FastAPI shines with async I/O workloads, which this isn't. For production scale I'd swap Flask's dev server for Gunicorn (which is already in the Procfile) behind an Nginx reverse proxy.

Q. Your MAPE is ~1.3% — isn't that suspiciously low?

A. No. Daily closes barely move, so the persistence baseline ("tomorrow = today") already achieves 1.34% MAPE. My model beats it by only 0.37%, which I surface on the dashboard. This is the honest state of daily-close forecasting from OHLCV alone.

Q. How do you know it's not overfitting?

A. Train MAE $\approx$ 1.95, test MAE $\approx$ 3.61 — a $\sim 2\times$ gap, which is mild and healthy. Catastrophic overfitting would be train 0.1 vs test 5. I also use TimeSeriesSplit (not random shuffle) during grid search so there's no look-ahead leakage.

Q. Why predict returns instead of the close price directly?

A. Trees can't extrapolate outside training range. If my training data tops at \$3000 and test data reaches \$4500, a tree-based predictor caps at \$3000. Predicting returns (a stationary target) and reconstructing $\text{close} = \text{last_close} \times (1 + \text{predicted_return})$ sidesteps the extrapolation problem entirely.

Q. What does "scale-free features" mean and why does it matter?

A. Every input feature is a ratio or return, so the numeric range is comparable across tickers regardless of price level. That's what lets a single model generalise from ^GSPC at 7,000 to NVDA at 200. Without this, I'd need one model per ticker.

Q. Why both RF and LSTM? Which wins?

A. RF is a strong tabular baseline and gives free feature importances. LSTM captures 20-day sequential context. On this data they tie (1.34% vs 1.38% MAPE). That's itself a finding — recurrent structure doesn't always beat tree ensembles on small temporal data.

Q. How does sentiment enter the model?

A. VADER scores a headline to a compound value in $[-1, +1]$. The predict endpoint takes an optional headline, scores it, and overrides the sentiment_score feature for the latest bar. During training I use a proxy derived from prior-day returns because I don't have a dated news archive spanning 15 tickers $\times$ 7 years. The pipeline is honest about this in the README. Sentiment ranks #8 out of 18 in the tuned RF feature importances — measurable but not dominant.

Q. What's the "MAPE reduced by 2.34%" number?

A. I trained a default RandomForestRegressor (no tuning) and a tuned one. The tuned test MAPE is 2.34% lower than the default. Hyperparameters picked via TimeSeriesSplit(4) grid search on validation MAPE. This is the defensible version of the resume's "reduced MAPE via tuning" claim.

Q. How did you handle Yahoo Finance rate limiting?

A. yfinance's default requests session got 429'd during bulk downloads. Fix: curl_cffi with Chrome TLS fingerprint (impersonate='chrome'). To Yahoo it looks like a regular browser. Same session is reused at inference time. On Render (data-center IP), live fetch sometimes still fails, so I ship CSV snapshots as a fallback.

Q. Why isn't the LSTM running in production?

A. Render's free tier has 512 MB RAM. TensorFlow needs ~400 MB at runtime and is loaded before anything else can run. Once you add Flask, pandas, numpy and sklearn, the pod OOMs. In production I deploy RF-only via requirements-prod.txt and LOAD_LSTM=false. The LSTM code and weights live in the repo for code review. Real industrial deployments routinely ship the lightweight model for the same reason.

Q. What's the difference between R^2 on close (0.9999) and on returns (0.005)?

A. This is the autocorrelation trap. $\text{Close}[t+1] \approx \text{close}[t]$, so any model that predicts close close to yesterday gets 99.99% R^2 . Returns are near-random, so R^2 near 0 tells the honest story. I report both on the dashboard to pre-empt the "why 99% accuracy?" question.

Q. How is the training/test split structured?

A. 80/20 within each ticker by date. First 5.6 years per ticker train, last 1.4 years per ticker test. No shuffling — sequential order preserved. ~21,000 training rows, ~5,000 test rows total.

Q. What would you do next to improve this?

A. Three things. (1) Swap the sentiment proxy for a real dated news archive (NewsAPI or a Kaggle financial-news dataset). (2) Add a Redis cache so ticker OHLCV survives restarts. (3) Deploy LSTM on a paid tier or Hugging Face Spaces. Stretch goal: bootstrap confidence intervals from RF tree disagreement, and a trading-PnL evaluation with transaction costs.

Q. How is the Flask code organised?

A. app.py imports from src/predict.py. Predict.py loads a PredictionBundle dataclass at startup containing RF, LSTM, scaler, normalisation dict, and metrics. Each Flask route reads that bundle and never touches disk during request handling. Routes are thin — under 20 lines each.

Q. Can this actually be used for trading?

A. No. Directional accuracy of 53% looks like an edge, but even 1–2 bps of round-trip cost on daily rebalancing would wipe it. This is a portfolio piece demonstrating a working ML-to-Flask-to-cloud pipeline. It's not a trading system and I don't pitch it as one.

Q. What if the recruiter hits your live URL during the demo and it's slow?

A. Render's free tier spins down after 15 minutes idle — first request after that takes 30–60 seconds to wake the container. I tell the interviewer upfront: "free tier cold start, give it a sec." On a paid tier this goes away.

14 · Closing pitch / future work

What this project demonstrates about me.

- End-to-end ML engineering — data ingestion, feature engineering, training with proper time-aware validation, and productionisation.
- Full-stack capability — Python backend, HTML/JS frontend, cloud deployment with CI via GitHub → Render.
- Honest evaluation — persistence baseline, train/test gap reporting, and R^2 in both close and returns space so "99% accuracy" can't hide weak signal.
- Pragmatic engineering trade-offs — dropping LSTM from the production path to fit memory; keeping it in the repo for review.

Suggested resume bullet (defensible from these numbers).

Stock Market Prediction | Python · LSTM · Flask · scikit-learn · Render

- Built a cross-ticker next-day close forecasting service — live OHLCV via yfinance, 18 scale-free engineered features, VADER news sentiment, Random Forest + Keras LSTM ensemble trained on 15 tickers × 7 years (≈26K rows).
- Achieved 1.336% MAPE on held-out data; hyperparameter tuning reduced test MAPE by 2.34% vs untuned baseline. Benchmarked against persistence to measure real model lift.
- Deployed on Render with Gunicorn; dashboard shows per-ticker MAPE, lift, directional accuracy, R^2 , and RF feature importances.

Future enhancements roadmap.

- Real dated news integration (NewsAPI or Kaggle financial-news dataset).
- Redis-backed ticker cache + Celery for async background refresh.
- Docker image and Hugging Face Spaces deploy for the full LSTM path.
- Bootstrap confidence intervals from RF tree disagreement.
- Trading-PnL evaluation with transaction costs and slippage.

Repository — github.com/atishay01/stock-market-prediction • Live demo — stock-market-prediction-ggdg.onrender.com